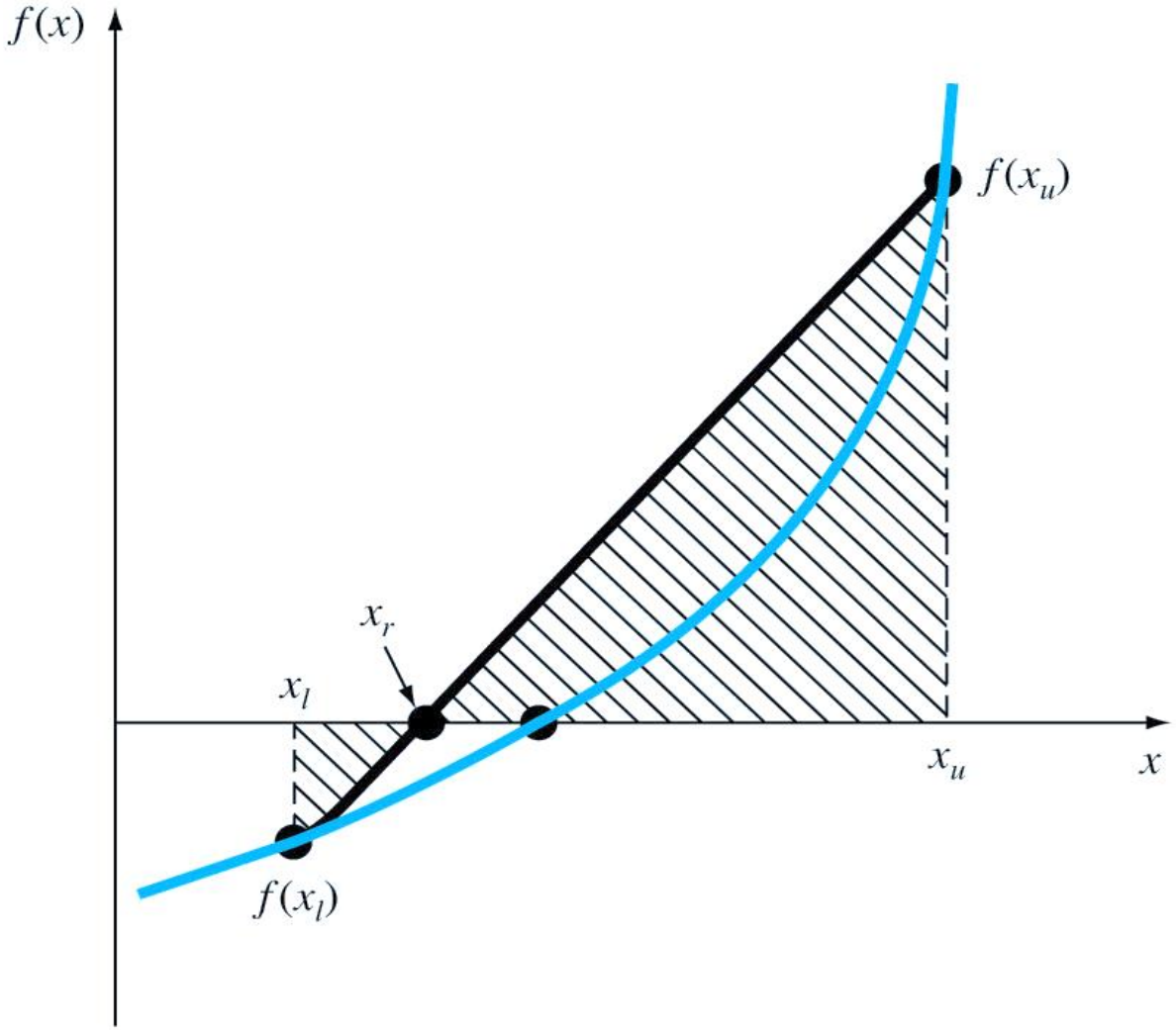
The False-Position Method (Regula-Falsi)

- If a real root is bounded by x_l and x_u of $f(x)=0$, then we can approximate the solution by doing a linear interpolation between the points $[x_l, f(x_l)]$ and $[x_u, f(x_u)]$ to find the x_r value such that $g(x_r)=0$, $g(x)$ is the linear approximation of $f(x)$.

== > Fig



Procedure

1. Find a pair of values of x, x_l and x_u such that

   $f_l = f(x_l) < 0$ and $f_u = f(x_u) > 0$.

2. Estimate the value of the root from the following formula

$$\frac{f(x_u) - f(x_r)}{x_u - x_r} = \frac{f(x_u) - f(x_l)}{x_u - x_l}$$

since $f(x_r) = 0$, $\frac{f_u}{x_u - x_r} = \frac{f_u - f_l}{x_u - x_l}$ => $x_r = x_u - \frac{f_u(x_u - x_l)}{f_u - f_l}$

finally, $$x_r = \frac{x_l f_u - x_u f_l}{f_u - f_l}$$

3. Use the new point to replace one of the original points, keeping the two points on opposite sides of the *x*-axis.

   a. If $f(x_r) < 0$ ==> root in the upper interval, then $x_l = x_r$ *and* $f_l = f(x_r)$

   b. If $f(x_r) > 0$ ==> root in the lower interval then $x_u = x_r$ *and* $f_u = f(x_r)$

   c. If $f(x_r) = 0$ then we have found the root and need go no further!

4. See if the new x_l and x_u are close enough for convergence to be declared. If they are not go back to step 2.

Why this method?

- Faster
- Always converges for a single root.

- Although false position often performs better than bisection, there are other cases where it does not. As in the following example $f(x) = x^{10} - 1$, there are certain cases where bisection yields superior results.

Advantages

1. It does not require the derivative calculation.
2. This method has first order rate of convergence i.e. it is linearly convergent. It always converges.

Disadvantages

Following are the limitations of regula falsi method

1. As it is trial and error method in some cases it may take large time span to calculate the correct root and thereby slowing down the process.
2. It is used to calculate only a single unknown in the equation.

Problem Solving

Use False-Position Method to determine the drag coefficient c_d needed so that an 80kg bungee jumper has a velocity of 36 m/s after 4 sec of free fall. Note: The acceleration of gravity is 9.81 m/s^2. Start with initial guesses of (c_d) x_l = 0.1 and x_u = 0.2 and iterate until the approximate relative error falls below 2%.

Solution: Velocity of falling bungee jumper is

$$v(c_d) = \sqrt{\frac{gm}{c_d}}\tanh\left(\sqrt{\frac{gc_d}{m}}t\right) \text{ or, } f(c_d) = \sqrt{\frac{gm}{c_d}}\tanh\left(\sqrt{\frac{gc_d}{m}}t\right) - v(c_d)$$

$$\text{or, } f(c_d) = \sqrt{\frac{9.81\times 80}{c_d}}\tanh\left(\sqrt{\frac{9.81\times c_d}{80}}4\right) - 36 \quad \ldots(1)$$

Problem Solving

Given, $c_d(\text{lower}) = x_l = 0.1$ and $c_d(\text{upper}) = x_u = 0.2$

Iteration – 1, $x_l = 0.1$ and $x_u = 0.2$

From equation 1, $f_l = f(0.1) = 0.86029$ and $f_u = f(0.2) = -1.19738$

$$x_r = \frac{0.1\times(-1.19738)-0.2\times0.8602906}{-1.19738-0.8602906} = 0.141808957$$

$f(x_r) = -0.03521093$, (-ve) therefore new $x_u = x_r = 0.141808957$

Iteration – 2, $x_l = 0.1$ and $x_u = 0.141808957$

$f_l = f(0.1) = 0.86029$ and $f_u = f(0.141808957) = -0.03521093$

$$x_r = \frac{0.1\times(-0.03521093)-0.141808957\times0.8602906}{-0.03521093-0.8602906} = 0.140165037$$

[Old approximation = old xr = xu] error = $((x_r - x_u)/x_r)* 100\% = 1.17\%$

Iteration		$f(c_d)$	x_r	error
1	x_u	-1.197380001		
2	x_l	0.860290613	0.141808957	1.17%
3	x_u	-0.035210927	0.140165037	
4	x_u	-0.000996447	0.140118569	
5	x_u	-2.81676E-05	0.140117256	
6	x_u	-7.96218E-07	0.140117219	

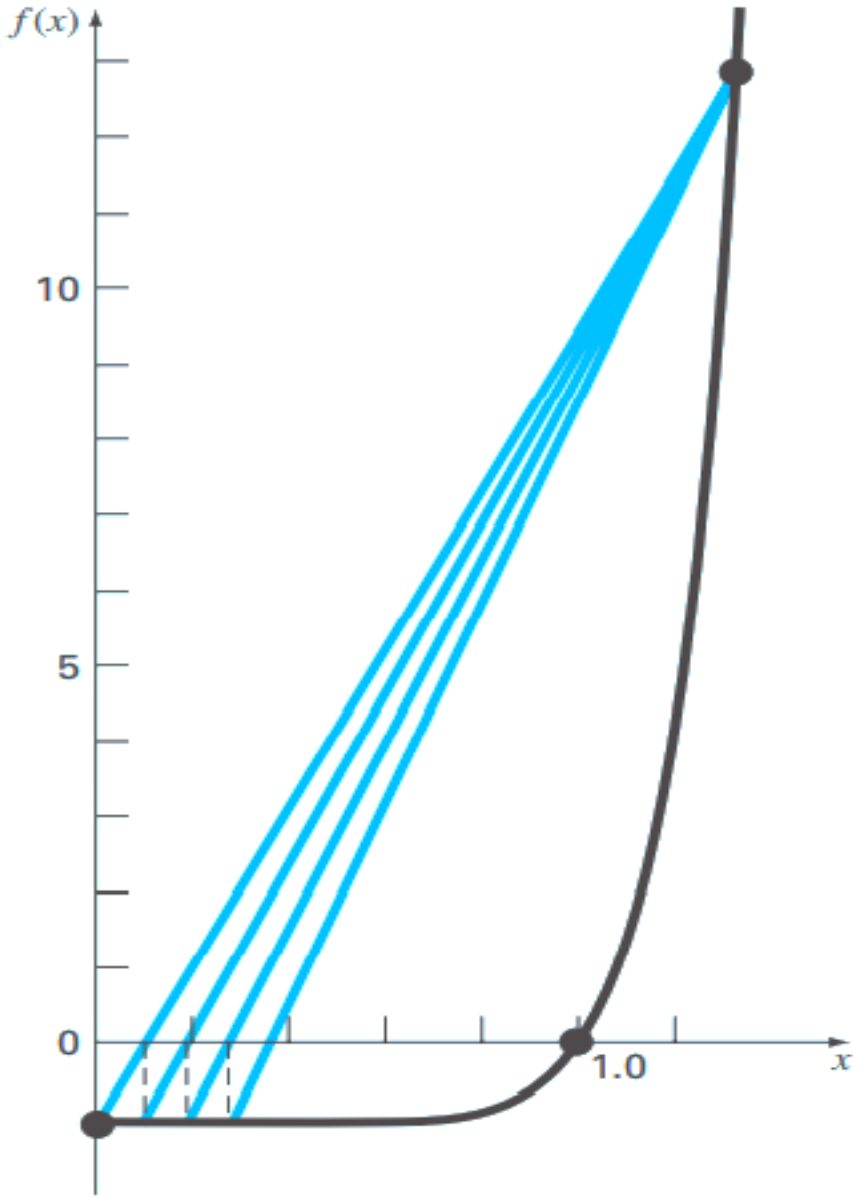
Comparison Bisection vs False Position

$$f(x) = 3x + \sin x - e^x = 0$$

$$x_0 = 0,\ x_1 = 1,\ f_0 \times f_1 < 0$$

k	x_k (Bisection)	f_k		x_k (False Position)	f_k
1	0.5			0.471	
2	0.25			0.372	
3	0.375			0.362	
4	0.3125			0.360	
5	0.34315	-0.042		0.360	2.93×10^{-5}

Certain Case



Plot of $f(x) = x^{10} - 1$, shows slow convergence of the false-position method

Bisection Method (Converge quicker)

Iteration	x_l	x_u	x_r	ε_a (%)	ε_t (%)
1	0	1.3	0.65		35
2	0.65	1.3	0.975	33.3	25
3	0.975	1.3	1.1375	14.3	13.8
4	0.975	1.1375	1.05625	7.7	5.6
5	0.975	1.05625	1.015625	4.0	1.6

False-position Method

Iteration	x_l	x_u	x_r	ε_a (%)	ε_t (%)
1	0	1.3	0.09430		90.6
2	0.09430	1.3	0.18176	48.1	81.8
3	0.18176	1.3	0.26287	30.9	73.7
4	0.26287	1.3	0.33811	22.3	66.2
5	0.33811	1.3	0.40788	17.1	59.2